Yalla: Estimating Muscle Growth, Maintenance, and Decline from Training Logs

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Abstract

Yalla estimates, per muscle and for the body overall, whether recent training is enough to grow, hold, or slowly lose muscle size. The estimate reads *training stimulus* from logged sets, loads, and reps; the app never sees the body itself. Two axes drive it: a *dose* axis (effort-weighted weekly sets against volume landmarks) and a *trend* axis (progressive overload measured as the better of a set trend and a load trend). Set effort is captured with a three-level control whose default is estimated from the load–rep relationship against the lifter's own best set. This document defines every quantity, the equations that combine them, the decision rules that produce a state, and the evidence each parameter rests on. It also states the model's assumptions, its validation status, and where it is deliberately conservative. The specification corresponds to the functions `growthStatus()`, `inferEffort()`, and `planForecast()` in `app.js`.

1 Introduction

Most training apps record what was lifted; few say what it means for muscle. Yalla estimates, per muscle and for the body overall, whether recent training is enough to grow, maintain, or slowly lose size, using only the sets, loads, and reps a user logs. This document specifies that model: the quantities it computes, the equations that combine them, the decision rules that produce a per-muscle verdict, and the evidence behind each parameter. The goal is an auditable feedback signal whose parameters trace to the resistance-training literature. Sections 3–11 define the model, Section 13 lists the evidence with a grade for each source, and Section 14 describes the checks that accompany the model in the absence of a body-composition study.

2 Background and related work

The model synthesises several strands of resistance-training research.

Volume and its counting. Schoenfeld, Ogborn & Krieger [13] established a graded dose–response between weekly set volume and hypertrophy, with each added set contributing on average about +0.37% of muscle across the studied range. The larger meta-regression of Pelland et al. [9] (67 studies, 2,058 subjects) confirmed rising-but-diminishing returns, found that counting indirect (secondary) sets at about half best fit the data, and found frequency largely neutral once weekly volume is matched. Yalla adopts fractional set counting (1) and derives its volume landmarks from a continuous dose–response curve (9) built on these findings.

Effort and proximity to failure. Sets far from failure contribute little; Refalo et al. [11] report that hypertrophy improves as sets approach failure, with roughly comparable outcomes across zero to four reps in reserve. Yalla scales each set by an effort factor (2) and, because reps in reserve cannot be read from load and reps alone, estimates it by default from the load–rep relationship [3] against the lifter's own best set.

Landmarks, maintenance, and decline. The MEV/MAV volume-landmark framework popularised by Israetel et al. [5] informs the landmark structure, though as a practitioner synthesis it carries the weakest evidence grade here. Bickel et al. [1] showed that built muscle is retained on a small fraction of the volume that built it, which sets the maintenance floor; Mujika & Padilla [6] documented the timeline over which an insufficient stimulus leads to loss, which motivates treating a sustained low dose as decline. Plotkin et al. [10] found that progressing load or reps drives near-identical growth, which is why the trend axis credits progress from either.

Positioning. These landmarks and principles are widely used as coaching guidance but are rarely turned into a per-log computed signal. Yalla operationalises them into a transparent, per-muscle estimate with stated uncertainty and a reproducible verification suite (Section 14).

3 Model overview

The unit of analysis is a muscle group g (chest, lats, quads, and so on). Training history is a set of logged entries; each entry e records one exercise in one session with a date d_e, a top-set load w_e (kg), top-set reps r_e, a set count n_e, an effort code $f_e \in \{0, 1, 2\}$, and a stored volume-load v_e.

Each muscle is placed in one of three states, combined from the two axes: *growing* (adequate dose and progressing), *holding* (a maintenance dose, or flat), and *under-stimulated* (below the volume needed to hold size). The dose axis answers whether there is enough stimulus this week; the trend axis answers whether the stimulus is rising, flat, or falling. A muscle grows only when both are satisfied: enough sets and a rising trend. This mirrors the requirement that hypertrophy needs both sufficient volume and progressive overload.

4 Volume attribution

4.1 Fractional set counting

Each exercise contributes to its primary muscle at full weight and to each secondary muscle at half. Writing the attribution weight of exercise x to muscle g as $\alpha(x, g)$:

$$\alpha(x,g) = \begin{cases} 1 & \text{if } g \text{ is the primary mover of } x, \\ 0.5 & \text{if } g \text{ is a secondary mover of } x, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

The 0.5 weight for indirect work is the fractional counting scheme that best fit the hypertrophy data in the Pelland et al. dose–response meta-regression [9].

4.2 Effort weighting

Only sets taken reasonably close to failure drive much growth, so each set is scaled by an effort factor $\varepsilon(f)$ before it counts:

$$\varepsilon(f) = \begin{cases} 0.5 & \text{Easy } (f{=}0,\ \geq 4 \text{ reps in reserve}), \\ 1 & \text{Hard } (f{=}1,\ \sim\!1\text{–}3 \text{ in reserve; the default}), \\ 1 & \text{Max } (f{=}2,\ 0\text{–}1 \text{ in reserve}). \end{cases} \tag{2}$$

Max carries no growth premium over Hard: hypertrophy is comparable across roughly 0–4 reps in reserve, and training to failure mainly adds fatigue [11]. Max still helps the load trend indirectly through the extra reps it buys. Entries with no recorded effort (all logs predating the feature) default to Hard, so $\varepsilon = 1$ and historical counts are unchanged.

4.3 Weekly effective sets per muscle

Time is bucketed into $N = 10$ rolling weeks; week index $k = 9$ is the current week. The effective weekly sets for muscle g in week k sum every contributing entry:

$$S_{g,k} = \sum_{e \in \text{week } k} n_e \, \varepsilon(f_e) \, \alpha(x_e, g). \tag{3}$$

This same effort-weighted, fractional count feeds the weekly-sets chart, the balance radar, and the "hard sets" ring, so every surface in the app reads consistently.

5 Effort inference

Proximity to failure cannot be recovered from load and reps alone: the same $100\,\text{kg} \times 8$ can be an all-out set or an easy one. So effort is a self-report. To keep friction to one optional tap, its default is estimated from the numbers and shown pre-selected; the lifter overrides only when the estimate is wrong.

5.1 Estimated one-rep max (anchor)

Set strength is summarised by the Epley one-rep-max estimate [3], with a rep denominator K (population default $K = 30$; Section 5.3 calibrates it per lifter):

$$\varphi(w, r) = w \left(1 + \frac{r}{K}\right). \tag{4}$$

The anchor for exercise x is the best estimate over the lifter's *prior* sessions for that exercise (the current, in-progress session is excluded):

$$A_x = \max_{e \in \text{hist}(x)} \varphi(w_e, r_e). \tag{5}$$

5.2 Reps in reserve and classification

Inverting (4) gives the reps a lifter should complete at load w before failure; subtracting the reps actually done estimates reps in reserve:

$$\text{RIR} = K \left(\frac{A_x}{w} - 1\right) - r, \tag{6}$$

evaluated on the session's hardest set for that exercise (the one with the largest φ). The estimate maps to an effort code:

$$f = \begin{cases} 2 \text{ (Max)} & \text{if RIR} \le 1, \\ 0 \text{ (Easy)} & \text{if RIR} \ge 4, \\ 1 \text{ (Hard)} & \text{otherwise.} \end{cases} \tag{7}$$

The anchor is recomputed on every estimate. Because it rolls, it adapts to deloads and off days and cannot lock in a wrong baseline, the failure mode of a one-time calibration. When there is no prior baseline for the lift, or the exercise carries no external load (bodyweight or timed holds), (6) is not evaluated and the default stays Hard.

5.3 Self-calibrated rep denominator

The denominator K in (4) varies by lifter and by lift. A set logged as Max is a near-true failure point, so two Max sets of the same exercise at different loads pin K: setting φ equal for both, $w_1(1 + r_1/K) = w_2(1 + r_2/K)$, which solves to

$$K = \frac{w_2 r_2 - w_1 r_1}{w_1 - w_2}. \tag{8}$$

Valid within-exercise pairs are pooled across the whole log, the median is taken and clamped to $[20, 40]$, and $K = 30$ is used until enough failure data exists. This grounds the effort estimate in the lifter's own data.

6 Volume landmarks

Four thresholds, in effective sets per muscle per week, define the dose axis. They follow the MEV/MAV framework and the dose–response meta-analyses.

Symbol (value)	Name	Meaning
MV (2)	Maintenance volume	Floor to hold existing size. Below it, a previously-trained muscle reads as under-stimulated.
MEV (6)	Minimum effective volume	Lower edge of the productive growth window; at or above it, an adequate dose is present.
target (10)	Practical target	Where most of the dose–response benefit has accrued; the balance-radar goal.
MAV (20)	Upper productive bound	Beyond it, added sets mostly add fatigue with little further growth.

A single set of landmarks is applied to every muscle. Real per-muscle landmarks differ, but one baseline keeps the output legible. The maintenance floor is deliberately low: built muscle is held on very little (as little as roughly one-ninth of the building volume in young trained adults [1]), so a 2-set floor avoids flagging loss too early.

6.1 Continuous backbone

The four landmarks discretise a continuous relationship [13, 9]. We model the fraction of the attainable per-muscle growth stimulus at s effective sets per week as a saturating curve $\rho(s)$, calibrated so the practical target captures about 80% of what is attainable:

$$\rho(s) = 1 - e^{-s/s_0}, \qquad s_0 = \frac{\text{target}}{\ln 5} \approx 6.21. \tag{9}$$

The landmarks are read off this single curve: $\rho(\text{MV}) \approx 0.28$, $\rho(\text{MEV}) \approx 0.62$, $\rho(\text{target}) \approx 0.80$, and $\rho(\text{MAV}) \approx 0.96$. So MEV is the ≈ 0.6 knee, target the 0.8 point, and MAV the ≈ 0.95 near-plateau. This replaces four independent thresholds with one meta-analytic function; the growth signal also reports $\rho(D_g)$ as a per-muscle share of the attainable growth stimulus.

7 Temporal windows

Both axes compare a recent window against an earlier one, each a three-week mean over the 10-week series. For any weekly series X_g:

$$\text{recent}(X_g) = \frac{1}{3}\sum_{k=7}^{9} X_{g,k}, \qquad \text{earlier}(X_g) = \frac{1}{3}\sum_{k=4}^{6} X_{g,k}. \tag{10}$$

The three-week mean smooths single missed or heavy sessions. The recent window is weeks 8–10 (the last three), the earlier window is weeks 5–7; the oldest three weeks provide history for charts but do not enter the comparison. Because the window is recent, the signal responds within about three weeks of a genuine change in training.

8 The two axes

8.1 Dose

The current dose for muscle g is the recent mean of effective weekly sets:

$$D_g = \mathrm{recent}(S_g). \tag{11}$$

8.2 Trend: progressive overload two ways

Overload can come from more effective sets or from heavier load; either counts as progress [10, 12]. A set trend and a load trend are formed as recent-over-earlier ratios. The load series $L_{g,k}$ is the best estimated one-rep max reached in week k on exercises for which g is the primary mover:

$$T_g^S = \frac{\mathrm{recent}(S_g)}{\mathrm{earlier}(S_g)}, \qquad T_g^L = \frac{\mathrm{recent}(L_g)}{\mathrm{earlier}(L_g)}. \tag{12}$$

Each ratio is defined only when its earlier window is positive; otherwise it is treated as absent. The combined trend takes the better of the two:

$$\tau_g = \max\left(T_g^S,\, T_g^L\right), \tag{13}$$

with an absent ratio contributing 0 to the maximum, and τ_g itself absent only when both ratios are absent (a muscle just started, with no earlier baseline). Taking the maximum prevents classic strength progression (add load, drop reps) from being misread as a decline: raw tonnage would fall there while T^L rises.

An earlier version used tonnage, $V = w\,r\,n$, as the sole trend signal. Tonnage rewards total kilograms moved, so lighter high-rep work inflates it while heavier low-rep work deflates it; it is a poor proxy for stimulus, and (13) replaces it.

9 Per-muscle state

The state σ_g is a function of the dose D_g and the trend τ_g, evaluated top to bottom (first matching rule wins):

1. **Under-stimulated** if $D_g < \mathrm{MV}$ (below the maintenance floor of 2).
2. **Under-stimulated** if τ_g exists and $\tau_g < 0.75$ (training eased off sharply).
3. Otherwise, if $D_g \geq \mathrm{MEV}$ (adequate dose, ≥ 6): **growing** if τ_g is absent or $\tau_g \geq 1.08$; **holding** otherwise ($0.75 \leq \tau_g < 1.08$).
4. Otherwise (maintenance band, $2 \leq D_g < 6$): **growing** if $\tau_g \geq 1.15$ and $D_g \geq 5$ (climbing toward a growth dose); **holding** otherwise.

The asymmetry is intentional: a muscle grows only with both an adequate dose and a rising trend, but it can be flagged under-stimulated by a low dose alone. Holding is the wide middle band: enough volume to keep size, below the dose that adds it.

9.1 Borderline states

The inputs carry known noise: effort-weighted sets to about ± 1 set, and each trend ratio to about ± 0.06. When a muscle's dose or trend sits within that range of a decision boundary, it is flagged *borderline* and shown with a tilde to mark the verdict as provisional. The sensitivity analysis (Section 14) shows the growth-trend boundary is where this matters most.

10 Overall verdict

Let n_{grow}, n_{hold}, n_{shrink} count trained muscles in each state, and m be the number of trained muscles. Let τ_{all} be the recent-over-earlier ratio of whole-body weekly volume. The headline is:

1. **Need more data** if fewer than 3 weeks show any training, or $m = 0$.
2. **At risk** if $n_{\text{shrink}} > n_{\text{grow}}$ and $n_{\text{shrink}} \geq \lceil m/3 \rceil$.
3. **Growing** if $n_{\text{grow}} \geq \max(1,\ \text{round}(0.4\,m))$ and τ_{all} is absent or ≥ 0.95.
4. **Holding** otherwise.

The "at risk" headline requires under-stimulated muscles to both outnumber growing ones and reach a third of those trained, so a few neglected small muscles do not flip the whole read. It resolves to "at risk" chiefly when most of recent training is genuinely sparse.

11 Plan forecast

Separately from the history-based signal, the plan forecast projects a plan's *prescribed* weekly sets before any of it is logged. Prescribed sets per rotation are scaled to a weekly rate by how often the plan is run, capped so an ambitious frequency on a short rotation cannot over-promise:

$$\hat{s}_g = s_g \cdot \min\left(1.2,\ \frac{p}{W}\right), \tag{14}$$

where s_g is prescribed sets per rotation (primary plus half of secondary, as in (1)), p is planned sessions per week, and W is the number of distinct rotation workouts. Each trained muscle is then sorted into under-dosed ($<$ MV), growth window (MEV to 20), past-productive ($>$ 20), or maintenance, and the counts drive a plain-language outlook. The forecast promises no centimetres or kilograms; it reports the dose against the growth window and what to change.

12 Simulations

To show the model's behaviour, a lifter executes three standard plans over 24 weeks in a seeded simulation (`research/simulate.mjs`). Loads progress when a muscle's weekly dose reaches the growth window, hold at a maintenance dose, and drift down below it; the model reads each week from a rolling 10-week window. The plots use the app's own colours. Every number is reproducible: the same seed regenerates the data files these figures read.

Every figure here is a Monte Carlo summary: each scenario is rerun many times over its random execution noise (missed sessions, session-to-session variation), and the plots show the median with a 10–90% band. Figure 1 contrasts regular execution of the Upper/Lower plan (every session) with an irregular schedule (about 55% of sessions, at random). Regular execution settles high and steady with a tight band; the irregular schedule sits lower and its band is wide, because how much stimulus a haphazard week delivers depends heavily on which sessions happen to land.

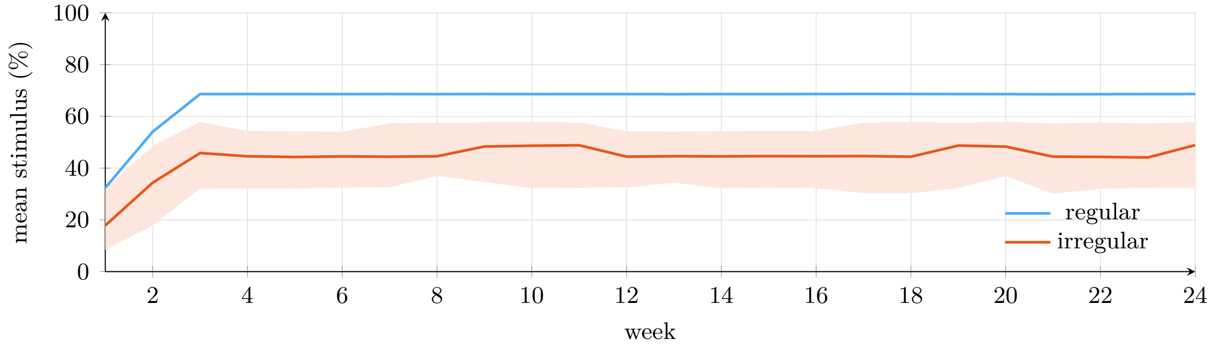

Figure 1: Regular (blue) versus irregular (orange) execution of the Upper/Lower plan: mean growth stimulus across nine muscles, median and 10–90% band over 80 runs. Regular is tight and high; irregular is lower and wide.

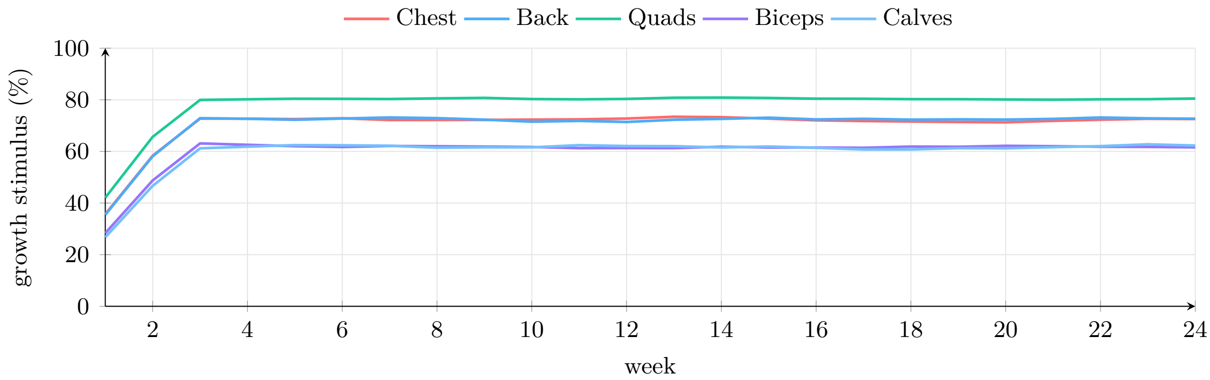

Figure 2: Per-muscle growth stimulus $\rho(D_g)$ on the Upper/Lower plan (regular), each muscle in its app colour.

Figure 2 traces five muscles on the Upper/Lower plan under regular execution, each in its app colour. All climb as the rolling window fills; the higher-volume muscles settle higher, the diminishing-returns shape of ρ.

Figure 3 compares the three plans under regular execution. More weekly volume per muscle lifts the whole-body stimulus, with the gap narrowing as the curve flattens.

Volume alone is not enough. Loads progress in these runs whenever a muscle's dose reaches the growth window, which is the progressive overload the app recommends; the trend axis is what rewards it. Figure 4 isolates that effect: one muscle held at a fixed adequate dose (10 sets/week) for 16 weeks, progressing at 2% per week versus repeating the same loads. The volume is identical, so the whole gap is progressive overload. Realised growth is $\rho(D)$ scaled by how far the trend sits in the growth band; without added load the trend flattens and growth settles far below its potential. The in-app forecast projects the same two levers forward.

12.1 Monte Carlo forecast

A point prediction of muscle gain would overstate what the model knows, so the in-app forecast is probabilistic. Each of 500 runs accumulates a weekly fractional gain

$$\Delta_t = b \cdot m \cdot \alpha \cdot \rho(D) \cdot \eta \cdot o \cdot e^{-t/32}, \tag{15}$$

where $\rho(D)$ is the dose–response of Section 4 [13, 9], η scales with proximity to failure [11], o is near 1 while progressing and drops without added load [10], b is a training-age base rate [2], α is an age factor that attenuates hypertrophy past about 30 [7], and m is the inter-individual

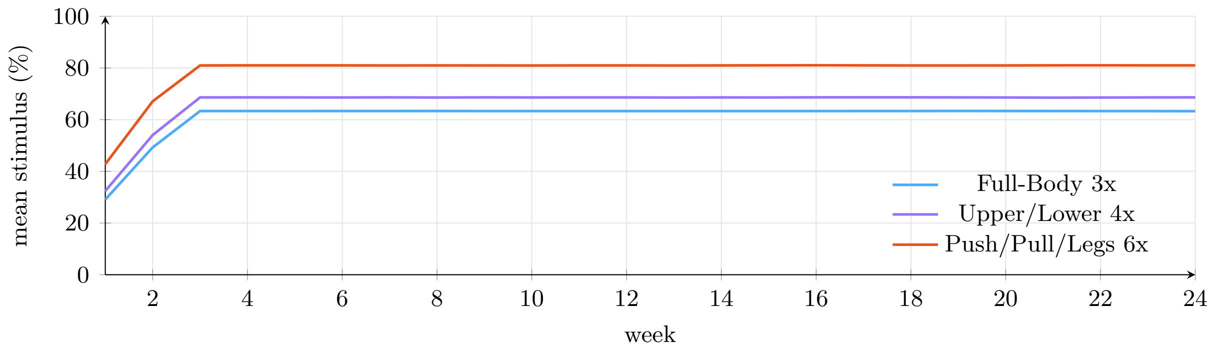

Figure 3: Mean growth stimulus across nine muscles for three standard plans, each executed regularly: median and 10–90% band over 80 runs. More weekly volume per muscle lifts the whole-body stimulus, with diminishing returns.

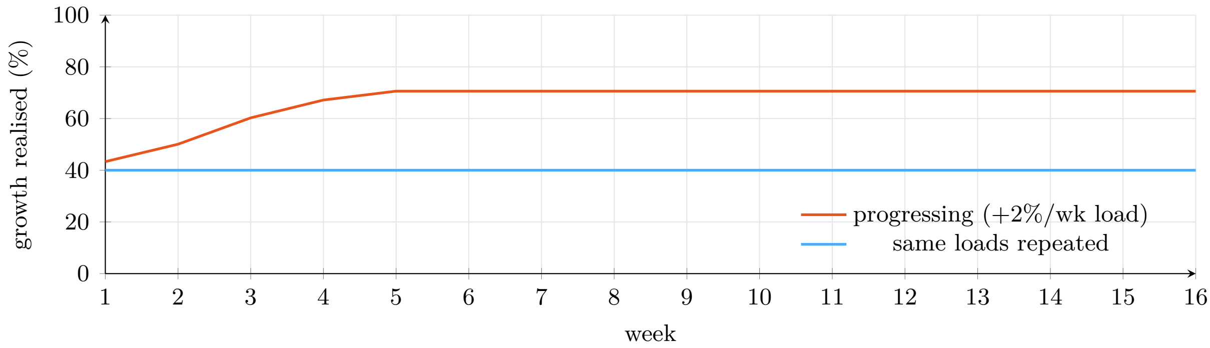

Figure 4: Progressive overload at a fixed adequate volume (10 sets/week): realised growth when loads climb 2% per week versus repeating the same loads. Identical volume; the difference is overload.

response multiplier, drawn lognormal to span the range Hubal et al. [4] observed (cross-sectional-area change from about −2% to +59%). Sex enters as a separate lever but is set near neutral: relative hypertrophy is comparable between sexes [8], so it informs absolute expectations rather than the percentage gain shown here. Below maintenance the increment turns negative (atrophy [6]), so a starved scenario trends down and the axis extends below zero. The first terms are well constrained by the literature; b is literature-informed with a wide prior, and m is the dominant source of spread, so the bands, not the median, carry the message. Figure 5 shows the median and 10–90% range for the chosen plan against the current pace; the same scenarios drive the forecast in the app.

Beyond the individual-response spread, Figure 6 shows how the median 16-week gain moves when each input is swung across a plausible range (the app draws the same tornado). Training experience and progression move it most; volume and age move it least over the ranges tested. This is a forecast-level companion to the threshold sensitivity of Section 14.

13 Scientific basis

Each parameter and mechanism traces to a specific source. Effect sizes are context-dependent; the model uses the literature to set its structure and thresholds, and promises no outcomes. Grades are GRADE-style ratings of the underlying evidence: High (meta-analyses of RCTs), Moderate (individual RCTs or focused reviews), Low (practitioner framework). A Low grade flags a parameter that rests on weaker evidence and is the first candidate for revision.

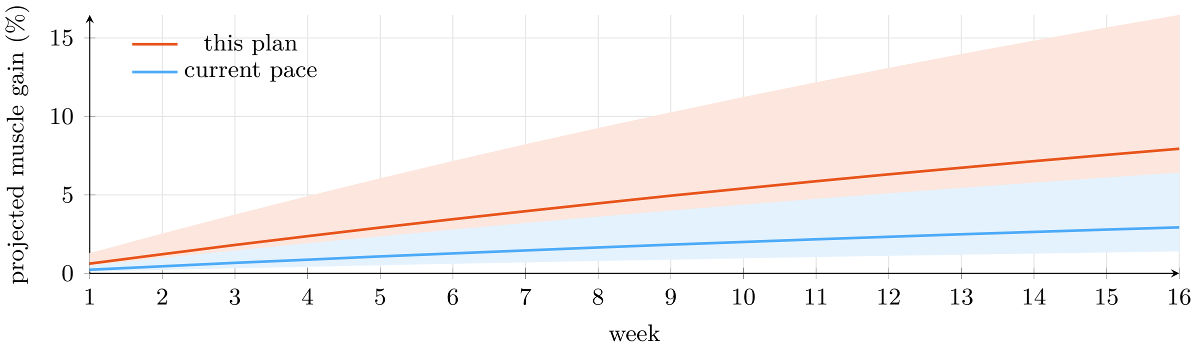

Figure 5: Monte Carlo muscle-gain forecast over 16 weeks (median lines, 10–90% bands) for a novice: keeping to an adequate, progressing plan versus a lower, stalling current pace. The wide bands are chiefly inter-individual variability [4].

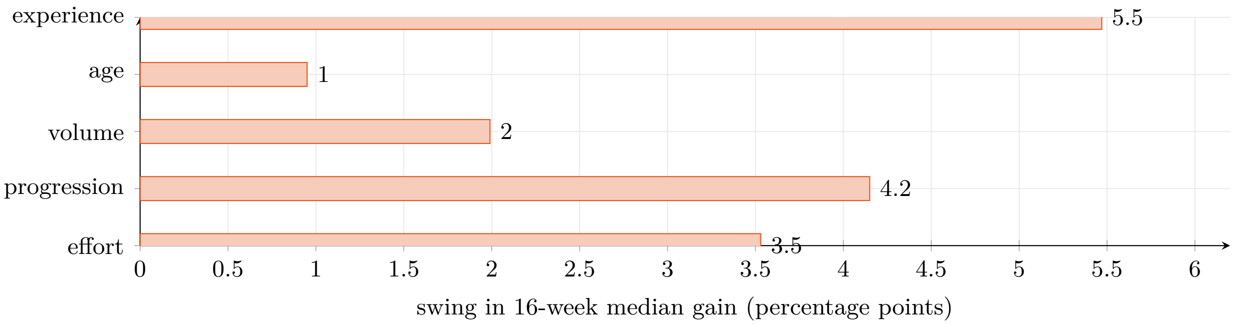

Figure 6: Parameter sensitivity of the Monte Carlo forecast: how far the median 16-week gain swings as each input moves across a plausible range, around a novice plan baseline.

Mechanism / parameter	**Basis**	**Grade**
Graded volume dose–response; ∼10 sets as a practical target	Schoenfeld, Ogborn & Krieger [13]: each added weekly set ≈ +0.37% muscle, benefit rising across the studied range.	High
Fractional (0.5) counting of indirect sets; diminishing returns; frequency neutral at equal volume	Pelland et al. [9] (peer-reviewed; 67 studies, 2,058 subjects).	High
Continuous backbone $\rho(s)$ and landmark derivation	Derived from the dose–response curves above [13, 9].	High
Effort factor ε; hypertrophy comparable across ∼0–4 RIR, dropping off when easier	Refalo et al. [11], systematic review and meta-analysis of proximity-to-failure.	High
MEV/MAV landmark structure (∼10 to 20 sets)	Israetel et al. [5], volume-landmarks framework.	Low
Maintenance floor MV; size held on ∼1/3 (older) to ∼1/9 (young) of building volume	Bickel, Cross & Bamman [1] (RCT, 16-wk build plus 32-wk reduced dose).	Moderate
Decline below maintenance stimulus; ∼2–4-week strength retention, then loss	Mujika & Padilla [6] detraining review; cross-sectional-area loss from ∼2–3 weeks of cessation.	Moderate
Overload via added load *or* reps; the trend tracks both	Plotkin et al. [10] (RCT; load- and rep-progression gave near-identical growth); mechanism in Schoenfeld [12].	Moderate
Epley one-rep-max φ; self-calibrated denominator K	Standard load–rep relationship [3]; K fitted from the lifter's own Max sets (8).	Method
Monte Carlo forecast: training-age base rate	Damas et al. [2]; hypertrophy rate depends on training status.	Moderate
Monte Carlo forecast: inter-individual response spread	Hubal et al. [4]9(n=585); 12-wk CSA change ≈ −2% to +59%.	High
Forecast age factor	Peterson et al. [7]; lean-mass gains attenuate with age.	Moderate

14 Verification

No body-composition study validates the model (see the limitations that follow). Three reproducible checks accompany it (the `research/` folder in the repository), each raising rigor without new measurement.

- **Known-answer tests.** Canonical scenarios — a returning lifter, a deload, easy versus hard sets, a hard stop — paired with domain-expected verdicts, so miscalibration or later drift is caught in code.
- **Sensitivity analysis.** Perturbing each threshold by $\pm 10\%$ over a synthetic population shows the growth-trend cutoff is the most load-bearing parameter (about a quarter of muscles change state), while the maintenance floor and sharp-drop cutoff are nearly cosmetic. This is why the borderline band sits on the trend boundary.
- **Predictive-coherence backtest.** On a training log, muscles labelled growing should precede larger subsequent gains in estimated 1RM than those labelled holding or under-stimulated. Strength change is only a proxy for hypertrophy; a body-composition study remains the standard these checks cannot replace.

15 Assumptions and limitations

- **Validation status.** This is a mechanistic model, grounded in the literature and internally consistent. It has no validation against direct body-composition measurements (DXA, ultrasound, or MRI): its thresholds are derived from published dose–response findings, none are fitted to an outcome dataset, and it claims no predictive-accuracy interval. The checks in Section 14 cover internal consistency and coherence; external accuracy stays out of scope.
- **A stimulus estimate.** Every output estimates the training stimulus from logs. The app cannot see the body; no measurement of muscle mass is implied.
- **Effort is self-reported.** Reps in reserve is not recoverable from load and reps alone; (6) sets only a default. Day-to-day max-rep variability ($\pm$1–3 reps) sits at the resolution of the effort bands, so the estimate is noisy near failure and the manual override remains the ground truth.
- **Top-set inference.** History stores each exercise's top set and its set count, without the reps of every set, so within-session rep drop-off is not yet used as an effort cue.
- **One landmark set for all muscles.** Chosen for legibility; real landmarks vary by muscle and training age.
- **Load trend needs external load.** Bodyweight and timed work do not contribute to T^L; those muscles rely on the set trend alone.
- **Residual tonnage use.** The overall verdict's τ_{all} still uses whole-body volume; the per-muscle trend does not.
- **Conservative by design.** The maintenance floor and the "at risk" threshold are set conservatively, biasing the model toward under-reporting muscle loss.

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A Symbol reference

Symbol	Definition
$g,\ x,\ e$	Muscle group, exercise, logged entry
$w,\ r,\ n,\ f$	Top-set load (kg), top-set reps, set count, effort code $\{0, 1, 2\}$
$\alpha(x, g)$	Muscle attribution weight: 1 primary, 0.5 secondary, 0 otherwise (1)
$\varepsilon(f)$	Effort factor: 0.5 / 1 / 1 for Easy / Hard / Max (2)
$S_{g,k}$	Effective weekly sets, muscle g, week k (3)
$\varphi(w, r),\ K$	Epley estimated one-rep max and its rep denominator (K=30 default, self-calibrated) (4), (8)
A_x	Anchor: best prior φ for exercise x (5)
$L_{g,k}$	Weekly peak φ on primary exercises for g
D_g	Recent dose: recent mean of S_g (11)
$\rho(s),\ s_0$	Dose–response stimulus fraction and its scale (9)
$T^S,\ T^L,\ \tau_g$	Set trend, load trend, combined trend (12)–(13)
σ_g	Per-muscle state (growing / holding / under-stimulated)
MV, MEV, target, MAV	Landmarks 2, 6, 10, 20 effective sets/muscle/week

B Parameter values

Every numeric constant in the model, its value, and where it enters. These are the values used in production (v121); the sensitivity analysis (Section 14) quantifies how much each one matters.

Parameter	Value	Where
Rolling window N	10 weeks	series length (3)
Recent / earlier windows	weeks 8–10 / weeks 5–7	(10)
Secondary-muscle weight	0.5	α (1)
Effort factors (Easy/Hard/Max)	0.5 / 1 / 1	ε (2)
RIR $\rightarrow$ effort bands	Max $\leq$1, Easy $\geq$4, else Hard	(7)
Epley denominator K	30 default, fitted then clamped to $[20, 40]$	(4), (8)
Dose–response scale s_0	target/ $\ln 5 \approx 6.21$	$\rho(s)$ (9)
Maintenance volume MV	2 effective sets/wk	state rule 1
Min. effective volume MEV	6 effective sets/wk	state rules 3–4
Practical target	10 effective sets/wk	radar goal; sets s_0
Upper bound MAV	20 effective sets/wk	plan forecast
Sharp-drop trend cutoff	$\tau_g < 0.75$	state rule 2
Growth trend cutoff	$\tau_g \geq 1.08$	state rule 3
Maintenance-band climb	$\tau_g \geq 1.15$ and $D_g \geq 5$	state rule 4
Borderline margins	$\pm$1 set; trend within 0.06 of $\{0.75, 0.92, 1.08\}$	Section 9.1
Overall “at risk”	$n_{\text{shrink}} > n_{\text{grow}}$ and $n_{\text{shrink}} \geq \lceil m/3 \rceil$	Section 10
Overall “growing”	$n_{\text{grow}} \geq \max(1, \text{round}(0.4m))$, τ_{all} absent or $\geq$0.95	Section 10
“Need more data”	fewer than 3 active weeks	Section 10
Plan frequency cap	$\min(1.2,\ p/W)$	(14)

Yalla training-stimulus model, v121. Estimates a training stimulus from logs; not medical or diagnostic advice, and not a measurement of muscle mass.